



## **Bryan Neumeister**

IEEE, IVPM, ABRE, ASCAP, PATC, APA, AES, BMI, NATAS

Court Certified: Military, Federal, State, Civil US District Court, Department of Justice,

US Grand Jury, State, Civil & Aviation Audio, Video & Photographic Forensic Expert

39 EMMY AWARDS for Technical Excellence

38 Years Professional Experience

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DARPA: Defense Advanced Research Projects Agency (classified)

Institute of Electrical and Electronics Engineers – Member

American Board of Recorded Evidence Standards

Technical work in 23 countries

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Over 70 National & International awards for technical excellence including Gold at the Cannes & Calgary film festivals for audio.

## **.jpeg metadata from Cell Phones and Cameras vs .pdf files**

Simply Put a .Jpeg is a photo format, a .pdf is not.

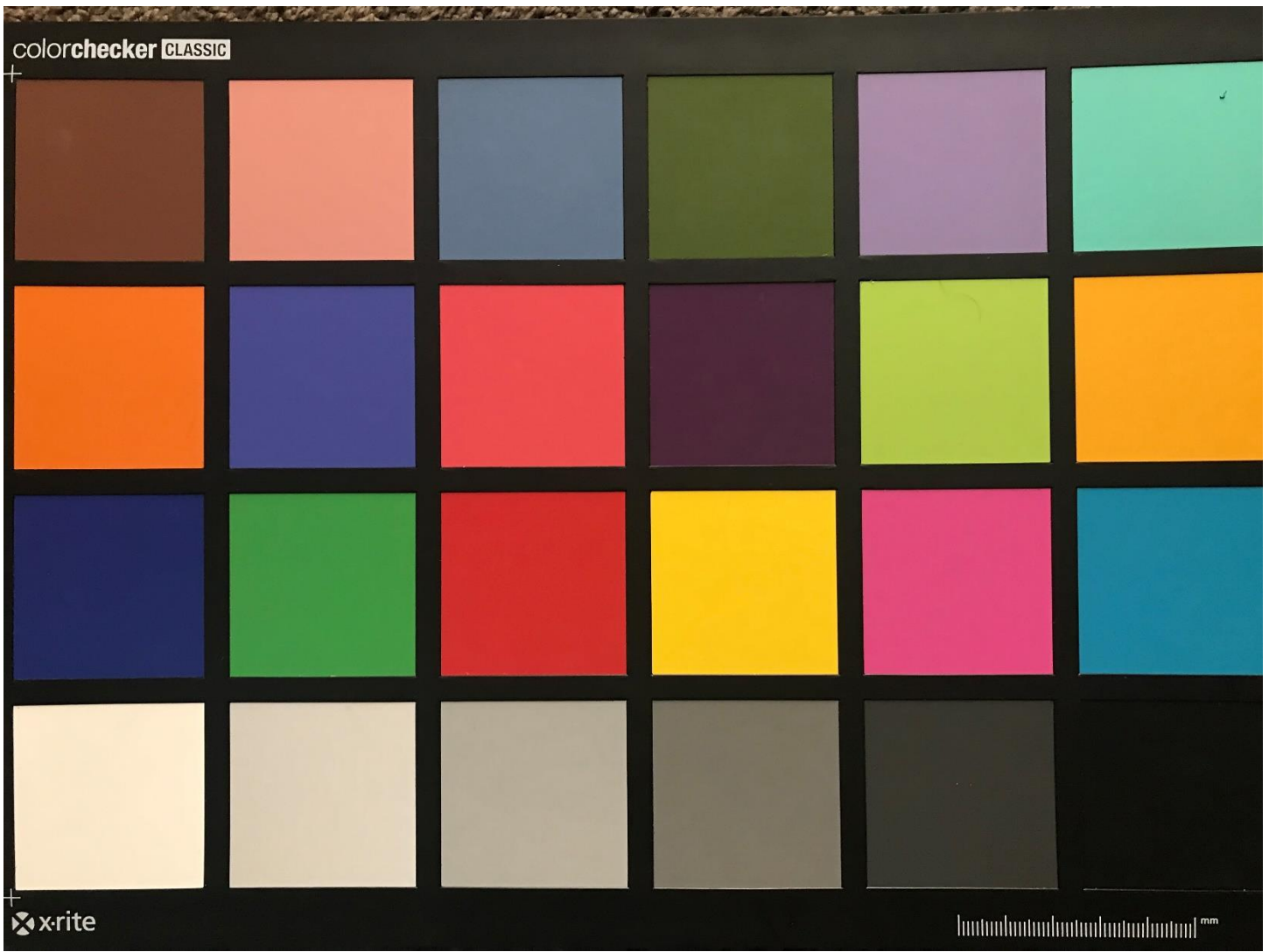
No Camera shoots in .pdf format.

One is a photo format - the other a document format.

When converting a photo to a .pdf all EXIF and metadata is lost including Time, Date, GPS location Data, type of camera, resolution, focal length, lens type etc...

## **Comparing the two:**

Below is a .jpeg of a Pantone set -up chart and its 6 full pages of metadata taken with an iPhone 7+. The most important camera & photo data is highlighted in Yellow, the location data in Red.



Filesize: [1455535] Bytes

Start Offset: 0x00000000

\*\*\* Marker: SOI (xFFD8) \*\*\*

OFFSET: 0x00000000

\*\*\* Marker: APP1 (xFFE1) \*\*\*

OFFSET: 0x00000002

Length = 9271

Identifier = [Exif]

Identifier TIFF = 0x[4D4D002A 00000008]

Endian = Motorola (big)

TAG Mark x002A = 0x002A

EXIF IFD0 @ Absolute 0x00000014

Dir Length = 0x000B

|                   |                                  |
|-------------------|----------------------------------|
| [Make             | ] = "Apple"                      |
| [Model            | ] = "iPhone 7 Plus"              |
| [Orientation      | ] = 6 = Row 0: right, Col 0: top |
| [XResolution      | ] = 72/1                         |
| [YResolution      | ] = 72/1                         |
| [ResolutionUnit   | ] = Inch                         |
| [Software         | ] = "11.2.6"                     |
| [DateTime         | ] = "2018:03:01 07:35:32"        |
| [YCbCrPositioning | ] = Centered                     |

```
[ExifOffset] = @ 0x00D2
[GPSOffset] = @ 0x06EC
Offset to Next IFD = 0x00000822
```

#### EXIF IFD1 @ Absolute 0x0000082E

```
Dir Length = 0x0006
```

```
[Compression] = JPEG
[XResolution] = 72/1
[YResolution] = 72/1
[ResolutionUnit] = Inch
[JpegIFOffset] = @ +0x0880 = @ 0x088C
[JpegIFByteCount] = 0x[00001BAD] / 7085
Offset to Next IFD = 0x00000000
```

#### EXIF SubIFD @ Absolute 0x000000DE

```
Dir Length = 0x0020
```

```
[ExposureTime] = 1/15 s
[FNumber] = F1.8
[ExposureProgram] = Normal program
[ISOSpeedRatings] = 50
[ExifVersion] = 02.21
[DateTimeOriginal] = "2018:03:01 07:35:32"
[DateTimeDigitized] = "2018:03:01 07:35:32"
[ComponentsConfiguration] = [Y Cb Cr .]
[ShutterSpeedValue] = 1805/462
[ApertureValue] = 2159/1273
[BrightnessValue] = 4031/2135
[ExposureBiasValue] = 0.00 eV
[MeteringMode] = Pattern
[Flash] = Flash did not fire
[FocalLength] = 4 mm
[SubjectLocation] = 2015, 1511, 2217, 1330
[MakerNote] = @ 0x02C0
[SubSecTimeOriginal] = "608"
[SubSecTimeDigitized] = "608"
[FlashPixVersion] = 01.00
[ColorSpace] = Uncalibrated
[ExifImageWidth] = 0x[00000FC0] / 4032
[ExifImageHeight] = 0x[00000BD0] / 3024
[SensingMethod] = One-chip color area sensor
[SceneType] = A directly photographed image
[ExposureMode] = Auto exposure
[WhiteBalance] = Auto white balance
[FocalLengthIn35mmFilm] = 28
[SceneCaptureType] = Standard
```

#### EXIF MakerIFD @ Absolute 0x000002CC

```
Makernote decode option not enabled.
```

#### EXIF GPSIFD @ Absolute 0x000006F8

```
Dir Length = 0x000F
```

```
[GPSLatitudeRef] = "N"
[GPSLatitude] = 33 deg 27' 2.640"
[GPSLongitudeRef] = "W"
[GPSLongitude] = 112 deg 4' 28.230"
[GPSAltitudeRef] = Above Sea Level
[GPSAltitude] = 330.880 m
[GPSTimeStamp] = 14:35:26.00
[GPSSpeedRef] = "km/h"
[GPSSpeed] = 0.000
[GPSImgDirectionRef] = "Magnetic direction"
[GPSImgDirection] = 62070/173
[GPSTestBearingRef] = "Magnetic direction"
[GPSTestBearing] = 62070/173
[GPSDateStamp] = "2018:03:01"
```

```
*** Marker: APP2 (xFFE2) ***
```

```

OFFSET: 0x0000243B
Length      = 564
Identifier   = [ICC_PROFILE]
ICC Profile:
  Marker Number = 1 of 1
  Profile Size      : 548 bytes
  Preferred CMM Type : 'appl' (0x6170706C)
  Profile Version    : 0.4.0.0 (0x04000000)
  Profile Device/Class : Display Device profile ('mntr' (0x6D6E7472))
  Data Colour Space   : rgbData ('RGB ' (0x52474220))
  Profile connection space (PCS) : 'XYZ ' (0x58595A20)
  Profile creation date : 2017-07-07 13:22:32
  Profile file signature : 'acsp' (0x61637370)
  Primary platform     : Apple Computer, Inc. ('APPL' (0x4150504C))
  Profile flags         : 0x00000000
  Profile flags         : > Profile not embedded
  Profile flags         : > Profile can't be used independently of embedded
  Device Manufacturer   : 'APPL' (0x4150504C)
  Device Model          : '....' (0x00000000)
  Device attributes     : 0x00000000_00000000
  Device attributes     : > Reflective
  Device attributes     : > Glossy
  Device attributes     : > Media polarity = negative
  Device attributes     : > Black & white media
  Rendering intent      : Perceptual
  Profile creator       : 'appl' (0x6170706C)
  Profile ID            : 0xCA1A9582_257F104D_389913D5_D1EA1582

```

\*\*\* Marker: APP10 (xFFEA) \*\*\*

```

OFFSET: 0x00002671
Length      = 764

```

\*\*\* Marker: DQT (xFFDB) \*\*\*

Define a Quantization Table.

```

OFFSET: 0x0000296F

```

Table length = 132

----

Precision=8 bits

Destination ID=0 (Luminance)

```

DQT, Row #0:  2  2  2  3  4  5  6  7
DQT, Row #1:  2  2  2  3  4  5  6  7
DQT, Row #2:  2  2  3  4  5  6  7  9
DQT, Row #3:  3  2  4  5  6  7  9 10
DQT, Row #4:  4  4  5  6  7  9 10 12
DQT, Row #5:  5  5  6  7  9 10 12 12
DQT, Row #6:  6  6  7  9 10 12 12 12
DQT, Row #7:  7  7  9 10 12 12 12 12

```

Approx quality factor = 93.45 (scaling=13.09 variance=12.88)

----

Precision=8 bits

Destination ID=1 (Chrominance)

```

DQT, Row #0:  2  2  3  5 12 12 12 12
DQT, Row #1:  2  3  3  8 12 12 12 12
DQT, Row #2:  3  3  7 12 12 12 12 12
DQT, Row #3:  5  8 12 12 12 12 12 12
DQT, Row #4: 12 12 12 12 12 12 12 12
DQT, Row #5: 12 12 12 12 12 12 12 12
DQT, Row #6: 12 12 12 12 12 12 12 12
DQT, Row #7: 12 12 12 12 12 12 12 12

```

Approx quality factor = 93.96 (scaling=12.07 variance=0.19)

\*\*\* Marker: DRI (Restart Interval) (xFFDD) \*\*\*

```

OFFSET: 0x000029F5

```

Length = 4

interval = 252

\*\*\* Marker: SOF0 (Baseline DCT) (xFFC0) \*\*\*

OFFSET: 0x000029FB

Frame header length = 17

Precision = 8

Number of Lines = 3024

Samples per Line = 4032

Image Size = 4032 x 3024

Raw Image Orientation = Landscape

Number of Img components = 3

Component[1]: ID=0x01, Samp Fac=0x22 (Subsamp 1 x 1), Quant Tbl Sel=0x00 (Lum: Y)

Component[2]: ID=0x02, Samp Fac=0x11 (Subsamp 2 x 2), Quant Tbl Sel=0x01 (Chrom: Cb)

Component[3]: ID=0x03, Samp Fac=0x11 (Subsamp 2 x 2), Quant Tbl Sel=0x01 (Chrom: Cr)

\*\*\* Marker: DHT (Define Huffman Table) (xFFC4) \*\*\*

OFFSET: 0x00002A0E

Huffman table length = 418

----

Destination ID = 0

Class = 0 (DC / Lossless Table)

Codes of length 01 bits (000 total):

Codes of length 02 bits (001 total): 00

Codes of length 03 bits (005 total): 01 02 03 04 05

Codes of length 04 bits (001 total): 06

Codes of length 05 bits (001 total): 07

Codes of length 06 bits (001 total): 08

Codes of length 07 bits (001 total): 09

Codes of length 08 bits (001 total): 0A

Codes of length 09 bits (001 total): 0B

Codes of length 10 bits (000 total):

Codes of length 11 bits (000 total):

Codes of length 12 bits (000 total):

Codes of length 13 bits (000 total):

Codes of length 14 bits (000 total):

Codes of length 15 bits (000 total):

Codes of length 16 bits (000 total):

Total number of codes: 012

----

Destination ID = 0

Class = 1 (AC Table)

Codes of length 01 bits (000 total):

Codes of length 02 bits (002 total): 01 02

Codes of length 03 bits (001 total): 03

Codes of length 04 bits (003 total): 00 04 11

Codes of length 05 bits (003 total): 05 12 21

Codes of length 06 bits (002 total): 31 41

Codes of length 07 bits (004 total): 06 13 51 61

Codes of length 08 bits (003 total): 07 22 71

Codes of length 09 bits (005 total): 14 32 81 91 A1

Codes of length 10 bits (005 total): 08 23 42 B1 C1

Codes of length 11 bits (004 total): 15 52 D1 F0

Codes of length 12 bits (004 total): 24 33 62 72

Codes of length 13 bits (000 total):

Codes of length 14 bits (000 total):

Codes of length 15 bits (001 total): 82

Codes of length 16 bits (125 total): 09 0A 16 17 18 19 1A 25 26 27 28 29 2A 34 35 36

37 38 39 3A 43 44 45 46 47 48 49 4A 53 54 55 56

57 58 59 5A 63 64 65 66 67 68 69 6A 73 74 75 76

77 78 79 7A 83 84 85 86 87 88 89 8A 92 93 94 95

96 97 98 99 9A A2 A3 A4 A5 A6 A7 A8 A9 AA B2 B3

B4 B5 B6 B7 B8 B9 BA C2 C3 C4 C5 C6 C7 C8 C9 CA

D2 D3 D4 D5 D6 D7 D8 D9 DA E1 E2 E3 E4 E5 E6 E7

E8 E9 EA F1 F2 F3 F4 F5 F6 F7 F8 F9 FA

Total number of codes: 162

----

Destination ID = 1

Class = 0 (DC / Lossless Table)

```

Codes of length 01 bits (000 total):
Codes of length 02 bits (003 total): 00 01 02
Codes of length 03 bits (001 total): 03
Codes of length 04 bits (001 total): 04
Codes of length 05 bits (001 total): 05
Codes of length 06 bits (001 total): 06
Codes of length 07 bits (001 total): 07
Codes of length 08 bits (001 total): 08
Codes of length 09 bits (001 total): 09
Codes of length 10 bits (001 total): 0A
Codes of length 11 bits (001 total): 0B
Codes of length 12 bits (000 total):
Codes of length 13 bits (000 total):
Codes of length 14 bits (000 total):
Codes of length 15 bits (000 total):
Codes of length 16 bits (000 total):
Total number of codes: 012

```

----

Destination ID = 1

Class = 1 (AC Table)

```

Codes of length 01 bits (000 total):
Codes of length 02 bits (002 total): 00 01
Codes of length 03 bits (001 total): 02
Codes of length 04 bits (002 total): 03 11
Codes of length 05 bits (004 total): 04 05 21 31
Codes of length 06 bits (004 total): 06 12 41 51
Codes of length 07 bits (003 total): 07 61 71
Codes of length 08 bits (004 total): 13 22 32 81
Codes of length 09 bits (007 total): 08 14 42 91 A1 B1 C1
Codes of length 10 bits (005 total): 09 23 33 52 F0
Codes of length 11 bits (004 total): 15 62 72 D1
Codes of length 12 bits (004 total): 0A 16 24 34
Codes of length 13 bits (000 total):
Codes of length 14 bits (001 total): E1
Codes of length 15 bits (002 total): 25 F1
Codes of length 16 bits (119 total): 17 18 19 1A 26 27 28 29 2A 35 36 37 38 39 3A 43
                                         44 45 46 47 48 49 4A 53 54 55 56 57 58 59 5A 63
                                         64 65 66 67 68 69 6A 73 74 75 76 77 78 79 7A 82
                                         83 84 85 86 87 88 89 8A 92 93 94 95 96 97 98 99
                                         9A A2 A3 A4 A5 A6 A7 A8 A9 AA B2 B3 B4 B5 B6 B7
                                         B8 B9 BA C2 C3 C4 C5 C6 C7 C8 C9 CA D2 D3 D4 D5
                                         D6 D7 D8 D9 DA E2 E3 E4 E5 E6 E7 E8 E9 EA F2 F3
                                         F4 F5 F6 F7 F8 F9 FA

```

Total number of codes: 162

\*\*\* Marker: SOS (Start of Scan) (xFFDA) \*\*\*

OFFSET: 0x00002BB2

Scan header length = 12

Number of img components = 3

Component[1]: selector=0x01, table=0 (DC), 0 (AC)

Component[2]: selector=0x02, table=1 (DC), 1 (AC)

Component[3]: selector=0x03, table=1 (DC), 1 (AC)

Spectral selection = 0 .. 63

Successive approximation = 0x00

\*\*\* Decoding SCAN Data \*\*\*

OFFSET: 0x00002BC0

Scan Decode Mode: No IDCT (DC only)

NOTE: Low-resolution DC component shown. Can decode full-res with [Options->Scan Segment->Full IDCT]

Scan Data encountered marker 0xFFD9 @ 0x001635AD.0

Compression stats:

Compression Ratio: 25.33:1  
Bits per pixel: 0.95:1

Huffman code histogram stats:

Huffman Table: (Dest ID: 0, Class: DC)

|                            |               |
|----------------------------|---------------|
| # codes of length 01 bits: | 0 ( 0%)       |
| # codes of length 02 bits: | 29108 ( 15%)  |
| # codes of length 03 bits: | 148814 ( 78%) |
| # codes of length 04 bits: | 2332 ( 1%)    |
| # codes of length 05 bits: | 2653 ( 1%)    |
| # codes of length 06 bits: | 2977 ( 2%)    |
| # codes of length 07 bits: | 3420 ( 2%)    |
| # codes of length 08 bits: | 1208 ( 1%)    |
| # codes of length 09 bits: | 0 ( 0%)       |
| # codes of length 10 bits: | 0 ( 0%)       |
| # codes of length 11 bits: | 0 ( 0%)       |
| # codes of length 12 bits: | 0 ( 0%)       |
| # codes of length 13 bits: | 0 ( 0%)       |
| # codes of length 14 bits: | 0 ( 0%)       |
| # codes of length 15 bits: | 0 ( 0%)       |
| # codes of length 16 bits: | 0 ( 0%)       |

Huffman Table: (Dest ID: 1, Class: DC)

|                            |              |
|----------------------------|--------------|
| # codes of length 01 bits: | 0 ( 0%)      |
| # codes of length 02 bits: | 81292 ( 85%) |
| # codes of length 03 bits: | 8495 ( 9%)   |
| # codes of length 04 bits: | 1486 ( 2%)   |
| # codes of length 05 bits: | 1437 ( 2%)   |
| # codes of length 06 bits: | 1152 ( 1%)   |
| # codes of length 07 bits: | 885 ( 1%)    |
| # codes of length 08 bits: | 509 ( 1%)    |
| # codes of length 09 bits: | 0 ( 0%)      |
| # codes of length 10 bits: | 0 ( 0%)      |
| # codes of length 11 bits: | 0 ( 0%)      |
| # codes of length 12 bits: | 0 ( 0%)      |
| # codes of length 13 bits: | 0 ( 0%)      |
| # codes of length 14 bits: | 0 ( 0%)      |
| # codes of length 15 bits: | 0 ( 0%)      |
| # codes of length 16 bits: | 0 ( 0%)      |

Huffman Table: (Dest ID: 0, Class: AC)

|                            |                |
|----------------------------|----------------|
| # codes of length 01 bits: | 0 ( 0%)        |
| # codes of length 02 bits: | 1050394 ( 52%) |
| # codes of length 03 bits: | 93435 ( 5%)    |
| # codes of length 04 bits: | 448683 ( 22%)  |
| # codes of length 05 bits: | 200784 ( 10%)  |
| # codes of length 06 bits: | 94687 ( 5%)    |
| # codes of length 07 bits: | 58092 ( 3%)    |
| # codes of length 08 bits: | 36507 ( 2%)    |
| # codes of length 09 bits: | 18730 ( 1%)    |
| # codes of length 10 bits: | 10707 ( 1%)    |
| # codes of length 11 bits: | 2428 ( 0%)     |
| # codes of length 12 bits: | 1318 ( 0%)     |
| # codes of length 13 bits: | 0 ( 0%)        |
| # codes of length 14 bits: | 0 ( 0%)        |
| # codes of length 15 bits: | 168 ( 0%)      |
| # codes of length 16 bits: | 2931 ( 0%)     |

Huffman Table: (Dest ID: 1, Class: AC)

|                            |               |
|----------------------------|---------------|
| # codes of length 01 bits: | 0 ( 0%)       |
| # codes of length 02 bits: | 120209 ( 73%) |
| # codes of length 03 bits: | 7448 ( 5%)    |
| # codes of length 04 bits: | 12102 ( 7%)   |
| # codes of length 05 bits: | 7887 ( 5%)    |
| # codes of length 06 bits: | 6481 ( 4%)    |
| # codes of length 07 bits: | 1276 ( 1%)    |
| # codes of length 08 bits: | 2361 ( 1%)    |

```

# codes of length 09 bits:      2417 ( 1%)
# codes of length 10 bits:     1229 ( 1%)
# codes of length 11 bits:     1033 ( 1%)
# codes of length 12 bits:      888 ( 1%)
# codes of length 13 bits:       0 ( 0%)
# codes of length 14 bits:      69 ( 0%)
# codes of length 15 bits:      30 ( 0%)
# codes of length 16 bits:     1020 ( 1%)

YCC clipping in DC:
  Y component: [<0=    0] [>255=  0]
  Cb component: [<0=    0] [>255=  0]
  Cr component: [<0=    0] [>255=  0]

RGB clipping in DC:
  R component: [<0=    0] [>255=  0]
  G component: [<0=    0] [>255=  0]
  B component: [<0=    0] [>255=  0]

Average Pixel Luminance (Y):
  Y=[ 98] (range: 0..255)

Brightest Pixel Search:
  YCC=[ 1000, -116,  62] RGB=[255,253,226] @ MCU[ 39,175]

Finished Decoding SCAN Data
  Number of RESTART markers decoded: 188
  Next position in scan buffer: Offset 0x001635AC.3

*** Marker: EOI (End of Image) (xFFD9) ***
  OFFSET: 0x001635AD

*** Searching Compression Signatures ***

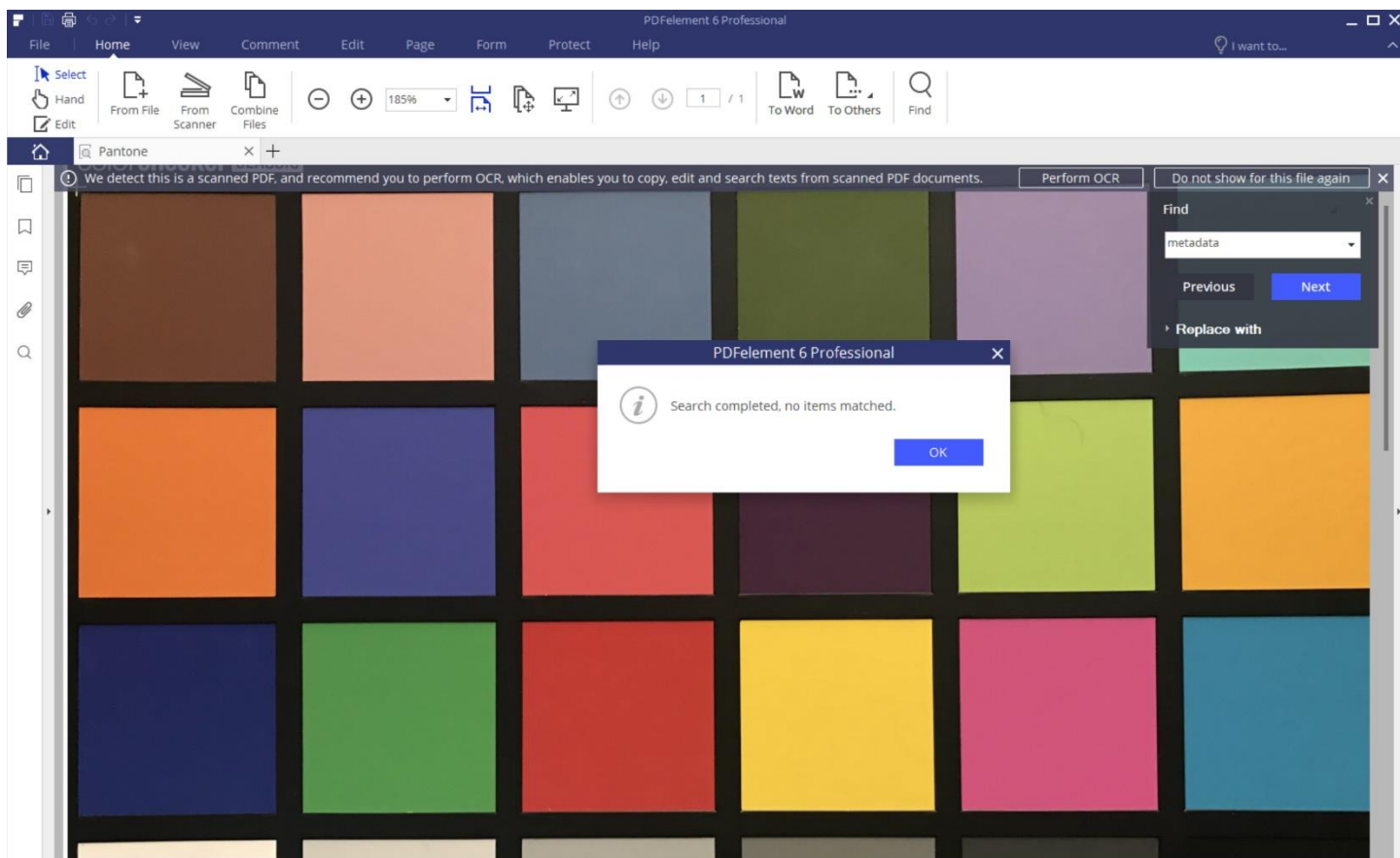
Signature:      012992123EA81665F11BD24AEA7540F8
Signature (Rotated): 01366131D922B72083CE7056DE281E17
File Offset:    0 bytes
Chroma subsampling: 2x2
EXIF Make/Model: OK   [Apple] [iPhone 7 Plus]
EXIF Makernotes: OK
EXIF Software:  OK   [11.2.6]

Searching Compression Signatures: (3347 built-in, 0 user(*) )

```

## This is the metadata from a .pdf file:

It had to be inserted as a screenshot because:  
a .pdf is not a photo format, it is a print document.



## All available metadata from .pdf

| General       |  | Security                          |  | Details |  | Previous Versions |  |
|---------------|--|-----------------------------------|--|---------|--|-------------------|--|
| Property      |  | Value                             |  |         |  |                   |  |
| File          |  |                                   |  |         |  |                   |  |
| Name          |  | Pantone .pdf                      |  |         |  |                   |  |
| Type          |  | PDF File                          |  |         |  |                   |  |
| Folder path   |  | C:\Users\Bruan Neumeister\Desktop |  |         |  |                   |  |
| Size          |  | 20.5 KB                           |  |         |  |                   |  |
| Date created  |  | 3/1/2018 7:53 AM                  |  |         |  |                   |  |
| Date modified |  | 3/1/2018 7:56 AM                  |  |         |  |                   |  |
| Attributes    |  | A                                 |  |         |  |                   |  |
| Owner         |  | USAFORENSIC-MAI\Bryan Neumeister  |  |         |  |                   |  |
| Computer      |  | USAFORENSIC-MAI (this PC)         |  |         |  |                   |  |

All the important data such as, Camera, GPS location, Time, RGB and any Chromatic information. **Noted in Yellow under the jpeg metadata of the .jpeg** is removed.

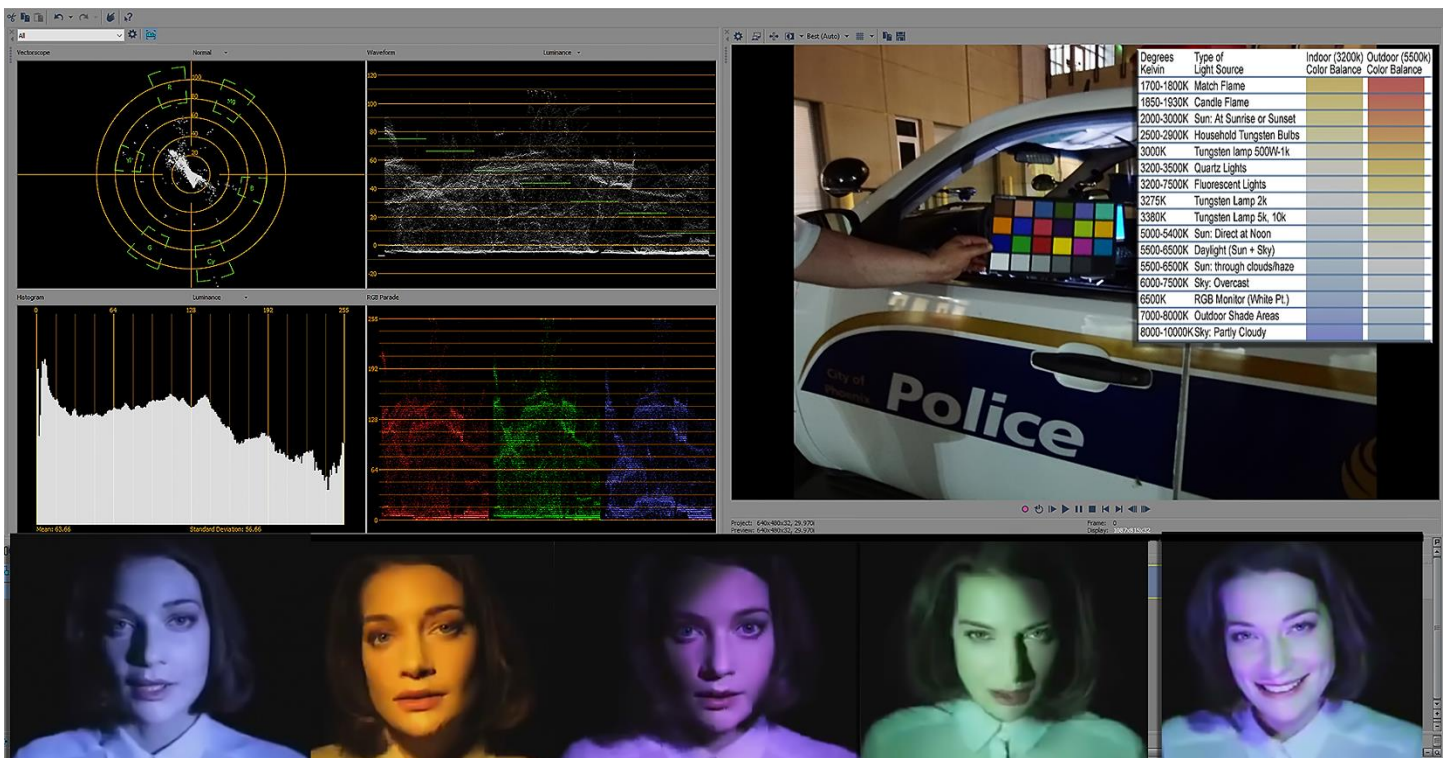
The color and luminance of the photograph will also be altered as the image will have been run through a computer that is likely not calibrated to accurately portray colors using a Pantone Chart along with a “Black, -18 Gray & White Card”, a Spyder screen-color analyzer and a Printer Calibrated to the Video card set-up.

Incorrect calibration can lead to inaccurate color renderings of flesh tones, wounds, bruises, vehicle colors under diverse types of light.

Light, measured in Degrees Kelvin, including *Daylight, Tungsten, Halogen, Neon, Sodium Vapor, Argon Vapor Led* and various light sources influences the way the human eye and cameras perceive color.

Below, a photo from a case I worked with the Phoenix Police Department, you can see the dramatic different lighting has on the female in the examples. *(the female at the bottom of the photo was not case related but is an example of how different lighting changes eye color, hair color and skin tone.)*

Note that a pantone chart was used in the photo to ensure color accuracy. The color and luminance was also measured on a VectorScope, Waveform Monitor, RGB Scope and Luminance Graph (upper left)



Once a photo is changed to a .pdf the color accuracy has been degraded significantly. Not only by the computer processing but printing of the document onto distinct types of paper. All papers contain chemicals that influence color replication. That's why a *calibrated* printer must be specifically set for the brand and type of paper being printed on.

Steganography (messaging or hidden code) is destroyed when converting images. It is very common for Drug cartels and Organized crime to encode .jpeg images with text or code written into the file.

It is an extremely straightforward process done with free Apps. All this information is lost during any type of conversion and often any type of compression.

Importantly: Any digital processing such as Photoshopping, will show up in the .jpg metadata..  
The metadata lists any alteration to the photo.

Below is the same pantone photo run through Photoshop for resizing and the first few lines of metadata.



```
EXIF IFD0 @ Absolute 0x00000014
Dir Length = 0x0010
[Make                      ] = "Apple"
[Model                     ] = "iPhone 7 Plus"
[Orientation                ] = 1 = Row 0: top, Col 0: left
[XResolution                ] = 720000/10000
[YResolution                ] = 720000/10000
[ResolutionUnit             ] = Inch
[Software                   ] = "Adobe Photoshop CS5 Windows"
[DateTime                   ] = "2018:03:01 08:30:35"
[YCbCrPositioning           ] = Centered
[ExifOffset                 ] = @ 0x0128
[GPSTimeOffset              ] = @ 0x0744
Offset to Next IFD = 0x0000087C
```

**For any scientific or legal use, chain of custody and photo authentication, the metadata must be examined, and the colors calibrated.**

## **For legal use:**

**Federal Rules of Evidence: 902(14) - 2017 (Daubert standards set in Dec 2017)**

**Rule 902(13):** **Certified Records Generated by an Electronic Process or System.** A record generated by an electronic process or system that produces an accurate result, as shown by a certification of a qualified person that complies with the certification requirements of Rule 902(11) or (12). The proponent must also meet the notice requirements of Rule 902(11).

**Rule 902(14):** Certified Data Copied from an Electronic Device, Storage Medium, or File. Data copied from an electronic device, storage medium, or file, if authenticated by a process of digital identification, as shown by a certification of a qualified person that complies with the certification requirements of Rule 902(11) or (12). The proponent also must meet the notice requirements of Rule 902(11).

While they may sound complicated, the intent of the new rules is actually very simple. Essentially, FRE 902(13) and (14) will allow parties to authenticate certain types of ESI without needing to offer any testimony as to foundation. Specifically, the following ESI will no longer need foundation-related testimony at trial: (1) A record generated by an electronic process or system that produces an accurate result, as shown by an appropriate certification<sup>4</sup>; or (2) data copied from an electronic device, storage medium, or electronic file, if authenticated by a process of digital identification, as demonstrated by an appropriate certification.<sup>5</sup>

The types of evidence that would fall under these rules could include, for example, GPS data, cell phone photos, text messages, and other electronic evidence, if the proponent introduced an authentication certificate, pursuant to FRE 902(13), showing that the ESI was obtained from systems that produced reliable results. FRE 902(14) will allow, among other things, self-authentication of forensic copies of web pages, text messages, or emails, certifying that they are the same as the originals.

Logistically, these rules would require the proponent of the ESI to present a **certification sufficient to establish the authenticity of the evidence**, reasonably far in advance of trial, at which point the opposing party would have to determine whether it could actually make a real challenge to the authenticity of the evidence.

This certification would need to be made by a qualified person (someone who would otherwise be able to testify at trial regarding authenticity) and, for the tech-savvy reader, would likely be performed by checking the **hash values for the original documents and the copies to ensure they are identical**, unless and until future technology provided new methods of identification. If the opposing party did not timely object to the certification, then no authenticating witness would be necessary at trial. The intent of the committee is to encourage parties to litigation to determine in advance of trial whether either intends to challenge the authenticity of any ESI, to appropriately tailor their trial preparation and streamline the trial itself.

In light of these amendments, it would behoove parties to litigation to make sure, in cases where large quantities of ESI could play a significant role at a potential trial, to collect such ESI using forensically sound methods, including employing specialists in appropriate cases and in any event ensuring that the methods used track the **hash values** of the documents. The clearer the records about what collection practices were utilized, and the more proactive counsel can be about giving notice to the opposing party and obtaining the appropriate certification in advance of trial, the more time and expense will be saved. Ideally, these amendments will result in smoother trial preparation, fewer witnesses, and shorter trials.

**Respectfully,  
Bryan Neumeister**

**Bryan Neumeister**

IEEE, IVPM, ABRE, ASCAP, PATC, APA, AES, BMI, NATAS

Court Certified: Military, Federal, State, Civil US District Court, Department of Justice,

US Grand Jury, State, Civil & Aviation Audio, Video & Photographic Forensic Expert

39 EMMY AWARDS for Technical Excellence

38 Years Professional Experience

USAForensic, Ilc. Court Certified Audio, Video, Computer, Cell Phone & Cell Tower Forensic Experts.

(602) 740-6128 : 44 W. Monroe St. 33<sup>rd</sup> floor, Phoenix, Az. 85003

[www.USAForensic.com](http://www.USAForensic.com)

United States District Courts: Contracted Technical Expert  
United States Grand Jury: Contracted Technical Expert  
US Department of Justice: Contracted Technical Expert  
United States ARMY - (SAMS Forensic contract with C.I.D.)  
Special Prosecutor's Office on Corruption - P.A.N.E.L. - US & Puerto Rico  
Audio Engineering Society- Member & Lecturer  
DARPA: Defense Advanced Research Projects Agency (classified)  
Institute of Electrical and Electronics Engineers – Member  
American Board of Recorded Evidence Standards  
Technical work in 23 countries  
Recipient of 39 EMMY AWARDS – National Association of Television Arts & Sciences for Technical Excellence  
Over 70 National & International awards for technical excellence including Gold at the Cannes & Calgary film festivals for audio.

### **Legal & Technical:**

38+ Years Professional Full-time Audio & Video Experience  
Professional Audio & Video Enhancement,  
Computer Drive Recovery,  
Cell Phone Forensics – Tower Analysis  
Federal & State Government Agencies Computer Drive Recovery  
DARPA: Defense Advanced Research Projects Agency (classified)  
Law Enforcement Agencies, National, State & Local  
8 Years Search & Helicopter Rescue (MCSO-DPS- Arizona)  
US Military – Army, Navy, Air Force, Marines & Coast Guard (SAMS contract)  
Los Angeles District Attorney's Office -NELOS  
Department of Defense (classified & non-classified)  
Beta Tester for Various Forensic Systems Manufacturers  
United States District Court system: Forensic Consultant  
United States Grand Jury system: Forensic Consultant  
U.S. Department of Justice: Forensic Consultant- Audio forensic work  
OPDS and Office of the Public Defender: Vendor number  
Work with US Army CID Prosecutors Office & Maricopa County Prosecutors Office  
Phoenix PD Homicide - Mesa PD Homicide- Case Video & Audio Evidence Enhancement  
Work with various Innocence Projects - Nation wide  
10 Years with NBC Television - Technical Video / Audio  
Federal, State, Aviation and Civil Law cases - Court Certified Technical Expert  
2 State of the Art Facilities. Over 75 custom Computers & Processors  
State of the Art HD Video and Audio Gear.  
Gear and Software Upgraded Continuously  
3D Sonographs, Spectrographs, Spectral 3D -HD Video & 192kHz - 64 bit Audio  
Expert Cell Phone Analysis (Oxygen) : Towers, Transmitters, Phones & Software  
Current D.O.E. Military Forensic High-Speed Video Expert  
Photo Enhancement using Clear-ID, Pixel Stacking & Frame averaging etc.  
Graduated Pi Sigma Alpha- California State University Northridge - 1980 (Political Science, emphasis on Journalism)  
Have worked in 23 Countries as a Technical Expert  
Published Nationally

### **Over 200+ Forensic Cases in the past 36 months:**

Non-Military court cases include: Federal, US District, Grand Jury, RICO, Homicide, Murder for Hire, Fraud, Robbery, Armed Robbery, Assault with a Deadly Weapon, Arson, Kidnapping, Bank Robbery, DUI, DWI, Civil Cases...etc. Currently working over a dozen Homicide and Capital cases nationwide. For Defense, Prosecution, Civil and Insurance carriers.

### **Large Cases (Small Sample):**

State of Arizona vs. Jodi Arias - In Trial, Audio Expert & Photographic Expert.  
US Government vs. Charles Keating - In Trial, Retained as Video Expert  
US Grand Jury: The Hope Steffy - Retained as Video Expert  
Government of India: Sri Nithyananda Swami - Retained as Video Expert  
US Government vs. Charles Keating III - In Trial, Retained as Video Expert  
State of Arizona vs. Steven DeMocker - Retained as Audio & Photographic Expert  
State of Arizona vs. Jeffery Martinson - Retained as Video Expert

United Kingdom (London, Barrister) V. Terone Williams - Retained as Audio Expert

State of Arizona vs. Jodi Arias – *Penalty Phase* - Computer Expert - *"During the first trial in 2013 and earlier in this trial, Mesa police experts testified that there was no porn and or viruses on the computer. Martinez had used that testimony to impeach Arias' claims that there were both, and it furthered his portrayal of Arias as a liar. But defense experts subsequently found porn and the viruses associated with the device. Mesa police have since admitted it was there all along."* –Gannett

### **Publications:**

Published Nationally

The Legal Investigator Magazine- Surveillance Systems - TV vs. Reality

High Tech, High Stakes Published in Expert Ease - National Forensic Publication.

Consultant to CPU Magazine as a Forensic Audio and Video Expert on Hardware and Software.

### **Books:**

*"Earthbound Misfit"* (Helicopter Rescue, TV Helicopters, Medivac & work with Law Enforcement agencies)

591 pages - *senior contributing author & consultant.*

*"The Secrets of the Blue Oyster Cult"* - Bryan Neumeister discussed as Blue Oyster Cult musician & composer on

*"Curse of the Hidden Mirror"* & *"Heaven Forbid"* Albums.

*"After You're Dead"* -Featured as Forensic expert *character*, "Bry", in #1 best selling series of Crime/Fiction books by author Cary Allen Stone.

*"S.E.D.S, The Journey Begins"* - Science Fiction. Character likeness and full name used in book as a scientist.

-Autor #1 bestselling Amazon author Cary Allen Stone.

### **National News Media: Forensic Analyst:**

**FOX Business Network:** Featured Forensic Expert 2018

**The Science Channel:** Six 1-hour Episodes on Video & Audio Forensics 2018

**Universal Studios/BBC & MSNBC:** Video Forensic Expert 2017: *Real, Fake or Unknown* TV series

Six 1-hour specials on analyzing viral videos

**Fox Business Network:** Featured Forensic Expert 2017

**CNN:** Jodi Arias: Audio & Photographic Forensic Expert (in Trial testimony)

**FOX News: Boston Bombings:** Forensic Surveillance Expert (analyst)

**CNN: Trayvon Martin / Zimmerman:** Audio Forensic Expert (analyst)

**NBC:** Forensic Surveillance Expert

### **Lectures:**

**2018 AACJ Annual Attorney Winter Conference (Facility)**

*Digital forensics: Audio, Video, Social Media, Cell Phones, Towers, Computers & Photography*

**2107 Audio Engineering Society:**

*Cellphone forensic audio re metadata, chain of custody, forensic acquisition of audio*

**2016 Federal Habeas Conference**

*Cellphone and Tower forensics*

**2016 Audio Engineering Society:**

*Topics: Multi-Pathing of Transmitted RF Signals, Forensic audio*

**2015 Inns of Court:**

*Topics: Cell Phone metadata, Audio & Video Forensics*

**2015 Audio Engineering Society:**

*Topics: Pareidolia, Dithering, Forensic Audio & Nyquist Theorem*

**2014 Audio Engineering Society & Conservatory of Recording Arts**

*Topics: Nyquist Theorem, Sample Dithering & Forensic Audio*

**Opening of the 2013 APDA (Arizona Public Defenders Conference)**

*Topics : Audio, Video, Photographic & Surveillance Forensics: 1.5 hour lecture*

**APDA Faculty Member** - 1,550+ Attorneys attended conference (June 26-28, 2013)

### **Television: National & International Broadcast:**

Recipient of **39 EMMY AWARDS** from The National Association of Television Arts and Sciences for Technical Excellence  
40+ additional National & International Awards.

Winner CANNES Film Festival (Gold) (Technical Audio Excellence)

Winner CALGARY Film Festival (Gold) (Technical Audio Excellence)

Worked for NPR & NBC Television News stations for 12+ Years.

Phi Sigma Alpha - National Political Science Honor Society – Lifetime Member.

Member: NAB, BMI, ASCAP, AES, ERA, NATAS

Winner: Film Advisory Board Gold Medal, Parents Choice Award

### **Film & Television Clients:**

NBC - ABC - CBS - PBS - BBC - FOX - DREAMWORKS - SCREEN GEMS FILMS - TOUCHSTONE FILMS - LUCASFILMS LTD. - LEVIS - COCA COLA - AMBLIN ENTERTAINMENT - PEPSI - VOLKSWAGEN - TOYOTA - CHEVROLET - NISSAN - AMERICAN EXPRESS - UNIVERSAL PICTURES - LORIMAR - TRIMARK FILMS - TRIMARK ENTERTAINMENT - GANNETT - A&E - DISCOVERY CHANNEL - FOX - DIAL SOAP - COORS BEER - BUDWEISER - MTV - PIZZA HUT - MOTOROLA - UNION CARBIDE - DISNEY - INTEL - MICROSOFT - VISA - MASTERCARD - ARMY - NAVY - AIR FORCE - MARINES - UNITED AIRLINES - CONTINENTAL AIRLINES - PARTNERS FILMS CANADA - PRTV - NATIONAL MEDIA - MOMENTUM FILMS – DFXTV - LOTTO/LOTTERY - GREYSTONE PICTURES - HISTORY CHANNEL – DISCOVERY CHANNEL - WARNER BROTHERS - DAIRY QUEEN - AT&T – MLB - NBA - NFL - SUPER BOWL XXX - ESPN - MAJOR LEAGUE BASEBALL – McDonald's - MCI - HARLEY DAVIDSON - STOUFFER'S- CARQUEST - MD HELICOPTERS - BOEING – “FUTUREWEAPONS”, “MYTHBUSTERS” {THE LUXOR - FLAMINGO - STRATOSPHERE - CESAR'S PALACE - MGM GRAND - NYNY - The HARD ROCK...CASINOS all in LAS VEGAS} - NUMEROUS ADVERTISING AGENCIES...

### **Certified Cell Phone Forensics:**

Latest Cellebrite Ultimate 4PC, Secure-View Forensics, Oxygen Detective & Analyst, APEX Laboratory and Field systems  
SV Striker box, PassWare Forensic, EnCase, etc.

D.A.R.T Tower Tracking and signal software- latest versions.

Certified Oxygen Forensic Analyst cell phone analyst 14,000+ phones (PATC)

PassWare Complete Forensic Decryption of 280+ encryption methods using 16 core liquid cooled systems

Recovery of deleted apps and data from over 14,000 cell phones.

All cell phone communications Apps, iCloud, iOS9, Android Lollipop, Windows OS etc. Password decryption & recovery.

DCode, Plist, SQLite Browser, Opanda, Kies, EnCase Extractor, EnCase readers, FTK Reader Stellar Phoenix Professional data recovery - 3 MSI Quad-Drive SSD field units for extraction

### **CELL TOWERS & CDR:**

D.A.R.T & 2018 Oxygen Detective Plotting

2016- Call Data Records & Towers (Lucent, Samsung, Nortel) PATC certification

AT&T, Verizon, Sprint, T-Mobile, Cricket, US Cellular, Tracfone, MetroPCS and subsidiaries...

Call Records Analysis – Switching analysis. Cloud, SMS, MMS E-Mail.

TOWER Dumps: Lucent, Samsung, Motorola, Ericsson etc. PATC 2016

GPS, NELOS, WiFi and Blue tooth tracking and E911-FCC mandated data location logs.

D.A.R.T. advanced and HTCI Mapping to verify if a call did hit a specific Tower Switcher.

Paperwork for legal requests for all of the above available on [www.USAForensic.com](http://www.USAForensic.com) .

### **Forensic Video & Surveillance Video Recovery/ Analysis**

DME Forensics DVR Examiner, iNPUT-ACE, Omnivor, D-plex Pro Forensic, D-Tective, ClearID Forensics and numerous forensic systems with over 200 clarification plug-ins

Latest computer and cell phone programs such as FTK, Autopsy, Cellebrite, Apex, Secure-view, Oxygen, EnCase...

Beta tester of Software & Hardware for leading Forensic Companies (Clear-ID, iZotope, DC8 Forensics etc)

Hard drive recovery for US Military (SAMs vendor) - Forensic drive analysis .

Recover damaged files, Deleted files, Changes in Registry & Meta-Data, Hash values, SHA2 etc

All Surveillance Systems as well as Computer Drives

Validate video pixel by pixel with iNPUT-ACE

Tampering or Keying detected.

Restore corrupted photographic or video files.  
Photo Enhancement using Clear-ID, Pixel Stacking & Frame averaging etc.  
Working with computers since 1977. 3 Field Extraction teams available.  
Network analysis, Tor tracing, Brute force & PassWare decryption etc.  
PassWare Forensic Decryption of 280+ encryption methods using 16 core liquid cooled systems  
Determining how files got onto a computer  
Validating dates of files/ changes to files via Sha2 HASH values.

### Computers

EnCase Ultimate, Autopsy, FTK and many other specialized tools:  
PassWare Forensic decryption. Custom machines.  
First code written in 1977 while attending Cal State University.  
Recovery using Tableau write-blockers USB 3.0 T8-R2 & T35ES-R2 Tableau & DME forensic Imaging.  
Beta tester of Software & Hardware for leading Forensic Companies (Clear-ID, iZotope, DC8 Forensics etc)  
Hard drive recovery for US Military (SAMs vendor) - Forensic drive analysis .  
Recover damaged files, Deleted files, Changes in Registry & Meta-Data, Hash values, SHA2 etc  
All Surveillance Systems as well as Computer Drives  
Restore corrupted photographic files.  
Photo Enhancement using Clear-ID, Pixel Stacking & Frame averaging etc.  
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PassWare Forensic Decryption of 280+ encryption methods using 16 core liquid cooled systems  
Determining how files got onto a computer  
Validating dates of files/ changes to files via Sha2 HASH values.  
Header Data verification

### Forensic Photographic work:

Photographic clarification, metadata validation  
Lux, Luminance & RGB analysis  
Film, Digital, thermal imaging and infra-red cameras  
Clear ID Forensics, iNPUT-Ace, Omnivore, Deplex  
Vectorscope, Waveform scope, Luminance and RGB scopes in Lap  
Pixel data analysis / Frame & Field level analysis  
Lens and lens-aberration analytics  
Lens MM relations to chip size and format  
Lens wide-angle distortions correction using metadata  
Deleted or Damaged Photo data memory cards recovered  
Cellphone camera metadata work.  
Geo data verification  
Dozens of forensic photo analytical plug-ins  
Stills lifted from video and clarified to 1200 dpi 8.5 x 11inch stills  
All stills printed on 80 lb postcard stock for legal use  
Studied from 1974-1979 with Dr Bill Wallner (co inventor of infra red photography)  
Cameras from high-end Hasselblad thru Panavision down to GoPro  
Ultra High-speed camerawork with Phantom-V cameras  
Cineflex & Cine-Alta.  
Helicopter mounts 14,500 hours aerial cinematography  
Have shot professionally in 23 countries  
Complete 4K 4:4:4 in lab.  
Pantone charting and color correction in field and lab  
Lux and Luminance lighting measurement in accident or crime scene cases.  
3-axis gyro-stabilized cameras  
All available camera mounts for GoPro 4K cameras for accident and test work.  
DOD and DOE military testing high speed and aerial photography.  
FAA licensed 4K Drone operator

Testified numerous times on photographic issues in Federal, Criminal, civil Military and aviation cases.  
Photo Work published internationally

### Field Cameras

For evidence gathering and accident recreation include Sony, HD & 4k, DJI Osmo, 3Axis stabilized 4k system, GoPro 4Ks all mounts and FAA Licensed 4k Drone. Fuji 3D Camera. Ultra slow-mo Phantom Cameras available. Meralli stabilization and re-speedR. Editing and all monitors are true 4k = 4086\*2160. Thermal Imaging. Thermal Video and still along with mixed format Thermal. Emmy Award winning crews for crime scene Recreation.

### Professional Sound & Music:

Over 20,000 voice tracks professionally worked with since 1980  
Federal, US District Court, State, US Military, Civil and Aviation cases to TV Network television, commercials, video games and films  
Sonographs, Spectrographs and 3D Audio Decibel and Frequency Mapping  
iZotope RX professional, 3D Spectral Audio, DC10 Forensics, Pro Tools  
Over 200 forensic plugins and programs running on 16 core liquid cooled computers  
Federal and State voice comparison cases  
Set Daubert standard for voice comparison in Arizona  
Phonetic breakdown of spoken words on spectrograph.  
Sonographs, Spectrographs and 3D Spectral audio graphs printed and embedded in forensic reports.  
Human Voice separated from background noises and ambience  
Tampering detection  
Metadata analysis  
Nyquist Theorem analysis  
Edit detection  
Dithering analysis  
Re-sampling analysis  
Audio background continuity  
Sine wave detection or embedded frequency (agency)  
AC power ground loop & harmonics removal.  
Background noise removal.  
Phase cancelation and removal of music from dialog  
Voice clarification, harmonic enhancement for muffled audio  
Removal of reverb (jail hallways or lecture halls etc)  
Removal of mic thumps and clicks that can be misidentified as edits.  
Transfer of tape to HQ digital medium: cassette, micro cassette, reel to reel  
Police radios split tracked  
32 bit audio running in 64 bit systems  
Conversion of any audio codec to another PROPERLY with correct dithering  
Time coding of audio for clients down to thousandths of a second for easy reference.  
IEEE engineering standards

Written & Recorded Music with *Blue Oyster Cult*, 24,000,000 (24 Million) albums sold - Gold & Platinum records & *Credence Clear Water Revival* recording artist Tom Fogerty (Movie Score).

-Mixed & Engineered for numerous top recording acts; Jazz, Rock & Classical.  
-Written music for hundreds of commercials including music cuts for Super Bowl XXX  
-Written Music/SFX for 5 popular Video Games.  
-Worked "Live" sound as sound mixer for concerts with crowds up to 35,000. *Challenging live mixes like: Chick Corea, Return to Forever, Ronnie Laws, Gerald Wilson's 24 piece Big Band etc.*

-Recently won another Emmy Award for best music on a Television series (Written & Performed).  
I have Mixed and Mastered hundreds of National, International & Regional Television Shows.

### **Aerial Photography / Rescue / Surveillance :**

Over 14,000+ Flight Hours of Jet Helicopter Photography – Film, HD Video, Research, Surveillance, Search & Rescue for Law Enforcement & Wildlife Rescue - Civilian Military Contractor.

Currently: Advisory Partner & Photographer - 4:4:4 Professional Film & TV Helicopter (*SaberCat HD*)

Pilot received the Harmon Flight Trophy from President Reagan at the White house. The Trophy is currently on display in the Smithsonian Air and Space Museum.

### **FAA Licensed: AERIAL 4K GPS-guided VIDEO DRONE**

FAA licensed 4K Video-drone, *Phantom Quadcopter*: Satellite-GPS controlled:

Owner & Operator); Mounted camera *GoPro Hero 4K Black Edition*

Thermal Imaging & Thermal Video Recording with image on image technology

Night Vision & Infra-Red

OSMO 3-axis stabilized 4k system, fully loaded.

### **Forensic Counter-Surveillance:**

Radio Frequency Response 10MHz ~ 10GHz (20+ GHz if needed)

Analog, Digital, WiFi, Cellular GSM/GPRS/EDGE/3G/4G signals all will set off alarms.

Detection Frequencies: GSM 880-915MHz, CDMA 824-849MHz, WCDMA (1920-1980MHz), and DCS (1710-1785MHz)

Digital 'Burst' Signal Detect for all GSM/3G/4G Trackers/SMS(Text) detection

Series and Parallel transmitters

Automobile GPS transmitters

Transmitting GPS trackers (only when GPS device is transmitting signal location)

Devices on internet phone and IP phone

Detect & Prevent:

Wire telephone tapping

Laser tapping

Recordings of a voice recorder, tape, digital and parabolic reflector using white noise & sibilance generators

40HZ 100-watt sine-wave generators to eliminate laser recording from windows

Thermal Imaging & Thermal Video Recording with image on image technology

1080P Sony Night Vision Cameras & Infra-Red Lighting

Osmo 3-axis stabilized 4k system

### **Weapons: Classified & Non-Classified:**

-Worked many homicide cases involving gunfire sound analysis, frequency, location & Triangulated acoustics.

-Worked with numerous surveillance videos to enhance shooting scenes for law enforcement and attorneys.

-Gunshots analyzed, compared & enhanced from: cell phones, land lines, surveillance equipment, 2-way communication, Police Duplex & Simplex, etc.

-Worked numerous cases requiring gunfire video enhancement from surveillance systems, cell phones and various recording devices.

-Have access to many renown civilian & military firearms experts for testimony as needed.

-Over 2-million rounds fired for high speed filming and forensic testing over 33 years. Work & have worked with -US Navy Seal Teams 1, 2 & 6, SOCOM, DEVGRU, 160th Night Stalkers. US Army Sniper teams. DOD & DOE.

-Additionally work with many US Government retained Military contractors.

-I currently have US Military SAMS / DUNS contract.

-Worked with numerous SWAT and TOU teams in actual field operations.

-Worked with LAPD, w/ Krav Maga training.

-MSCO Helicopter Rappel Trainer (1980s Tac-Ops-Unit) - MSCO Helicopter Search & Rescue = SKY12 & DPS

-Worked with Springfield Armory testing the initial launch of the entire XD Pistol line. 9 mm, .40, .357, .45

-Work often with DillonAero on the M-134 Mini-gun project (over 20 years working together).

-Analyze Muzzle flash and sound signature of McMillan Sniper rifles for US Army Snipers (over 10 years of work together).

-Air to Ground target acquisition and aerial sniper training documentation. Thousands of hours of aerial law enforcement, search & rescue & air to ground gunship work.

-Assigned to USS CVN70 Carl Vinson and Helicopter-Carrier USS New Orleans as civilian videographer.

-Have been a "go to" guy for live Weapons Photography & Sound Recording for The Military Channel, History Channel, MythBusters, Future Weapons, Discovery Channel, National Geographic, BBC and many others.

-Produced one of the largest selling automatic weapons DVDs ever made, "Firestorm in the Desert" &

-Produced one of the largest selling handgun training DVDs of all time with Rob Leatham, "Shooter Ready

-Currently: Civilian Technical Advisor on the board of CowTown Range & Studios, which is an extremely active 88 acre Training facility for SOCOM, Swat Teams, Navy Seal Teams, DEVGRU, Homeland Security and many Special Operations teams. CowTown Range is currently under DOD and various Agency contracts for range work, tactical training and weapons development.

### **Weapon recording analysis for cases includes**

Various rounds recorded over surveillance systems, cell phones and hand held video cameras.

Different types of pressure, loads, powders, barrel length and calibers recorded for analysis.

Cell phone cases include NyQuist Theorem frequencies relating to cell Phone FQ range, Sample Rate and Bit depth of recordings

Cell phone line noise and background audio of conversations recovered from 911 calls.

Dithering artifacts created during sample rate conversions.

Dithering harmonics relating to audio captured by Cellebrite or Oxygen Forensics, when transferred to 44.1-16 bit for distribution.

Cell call transmission, multi-pathing of signal, tower horn tracking v. switcher tracking, topographical anomalies of transmissions, transmission artifacts

Phase cancellation of recordings due to microphone set ups.

Distances of cell phone to firearm (FARO 130 system)

Shootings captured by police wire or boomerang unit

Sonographs and spectrographs along with 3D Spectral recording to narrow down gunshots to 1,000ths of a second.

Shooting weapons in range shoot houses or gun ranges recorded over cell phones.

Shots recorded over cell phones in cars, houses, open acoustic spaces

Ambiance and acoustics graphed and measured of gunshots on surveillance and cell phone

Certified Oxygen Cell Phone Forensics (PATC) Two Forensic Cell phone systems used 3 to 6 times a week on cases.

Working with Cell tower experts from Verizon currently on Federal case

Currently working with Court Certified retired US Navy Seal Team 6 and DEVGRU weapons expert, Craig Sawyer on two cases

Currently working with Law Enforcement and Prosecutor on a Homicide investigation involving shooting on surveillance.

Have worked with Phoenix Homicide quite often to assist with clarification of recorded audio on both homicide and cold cases.

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### **USA FORENSIC: AUDIO & VIDEO FORENSIC LABS**

USA Forensic, llc. (34 Years *Professional Experience*)

Certified Audio & Video Forensic Experts: Federal, Military, State, Civil & Aviation

United States District Courts - U.S. Grand Jury- DOJ, - DOE (Classified-Unclassified)

IEEE, IPVM, ACFE, AES, ABRE, APA, NATAS, BMI, ASCAP, PSA, ACFEi



(602) 740-6128

[www.USAForensic.com](http://www.USAForensic.com)

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Matt@USAForensic.com